

NC Medicaid Humira Biosimilar Transition Cost Analysis

Evaluating the Fiscal Impact of NC PDL Policy Change
Making Adalimumab Biosimilars Co-Preferred

Report Date: March 26, 2026

Data Sources: CMS State Drug Utilization Data CY2024, NADAC CY2025

NC PDL Effective January 1, 2026 (revised 03/18/2026)

Executive Summary

The North Carolina Division of Health Benefits revised its Preferred Drug List (PDL) effective January 1, 2026 to make multiple adalimumab biosimilars co-preferred alongside Humira in the Cytokine and CAM Antagonists therapeutic class. This analysis estimates the gross reimbursement impact of shifting Humira utilization to biosimilars at varying adoption rates using CMS State Drug Utilization Data (CY2024) and NADAC pricing (CY2025).

Key findings: NC Medicaid spent \$171.4 million (gross) on Humira in CY2024 across 44,310 units and 16,238 prescriptions. Biosimilar utilization was negligible (<0.4% of the adalimumab market). Co-preferred biosimilars carry NADAC prices 63-86% below Humira, with a blended average discount of 78%. At a 50% biosimilar adoption rate, estimated gross savings would be \$66.7 million annually (\$20.0 million state share at 30% FMAP).

IMPORTANT: These estimates reflect gross reimbursement costs only. Humira's lengthy market tenure (since 2002) generates substantial CPI-based Medicaid Drug Rebate Program (MDRP) rebates that significantly reduce its net cost. Biosimilars carry lower basic rebates (13% vs. 23.1%) and minimal CPI penalties. A complete net fiscal analysis requires confidential supplemental rebate data not available in public CMS datasets.

Key Metrics at a Glance

Annual Humira Gross Spend (CY2024)	\$171.4 million
State Share at 30% FMAP	\$51.4 million
Blended Biosimilar NADAC Discount	78%
Potential Gross Savings at 50% Shift	\$66.7M total / \$20.0M state
Potential Gross Savings at 75% Shift	\$100.1M total / \$30.0M state
Current Biosimilar Market Share (NC)	<0.4%

1. Policy Change: NC PDL Revision

The NC PDL (effective January 1, 2026, revised March 18, 2026) designates the following adalimumab biosimilars as co-preferred with Humira in the Cytokine and CAM Antagonists class. Prior authorization clinical criteria apply to all drugs in this class. Trial and failure (T/F) of one preferred drug is required.

Non-proprietary Name	Brand	Dosage Forms	PDL Status
adalimumab (Humira)	Humira	Pen / Syringe / Starter	Preferred
adalimumab-adbm	Cyltezo (generic)	Pen / Syringe	Preferred
adalimumab-afzb	Abrilada	Pen / Syringe	Preferred
adalimumab-bwwd	Hadlima	Syringe / PushTouch	Preferred
adalimumab-advo	Pyzchiva	Syringe / Vial	Preferred
adalimumab-ryvk	Simlandi	Autoinjector / Vial	Preferred

2. Baseline Utilization (CY2024)

North Carolina Medicaid utilization of adalimumab products in calendar year 2024, as reported in the CMS State Drug Utilization Data, is overwhelmingly concentrated in brand Humira. Biosimilar adoption has been minimal.

Humira Utilization Summary

Total Units Reimbursed	44,310
Total Prescriptions	16,238
Gross Total Reimbursed	\$171,393,345
Gross Medicaid Reimbursed	\$167,145,049
Avg Reimbursed Cost/Unit	\$3,868.05
Avg Reimbursed Cost/Rx	\$10,555.08

Existing Biosimilar Utilization (CY2024)

Product	Units	Rx	Total Reimbursed	Avg/Unit
Generic adalimumab	148	116	\$235,021	\$1,588
Hadlima	14	13	\$18,182	\$1,263
Hyrimoz	38	27	\$315,830	\$8,225

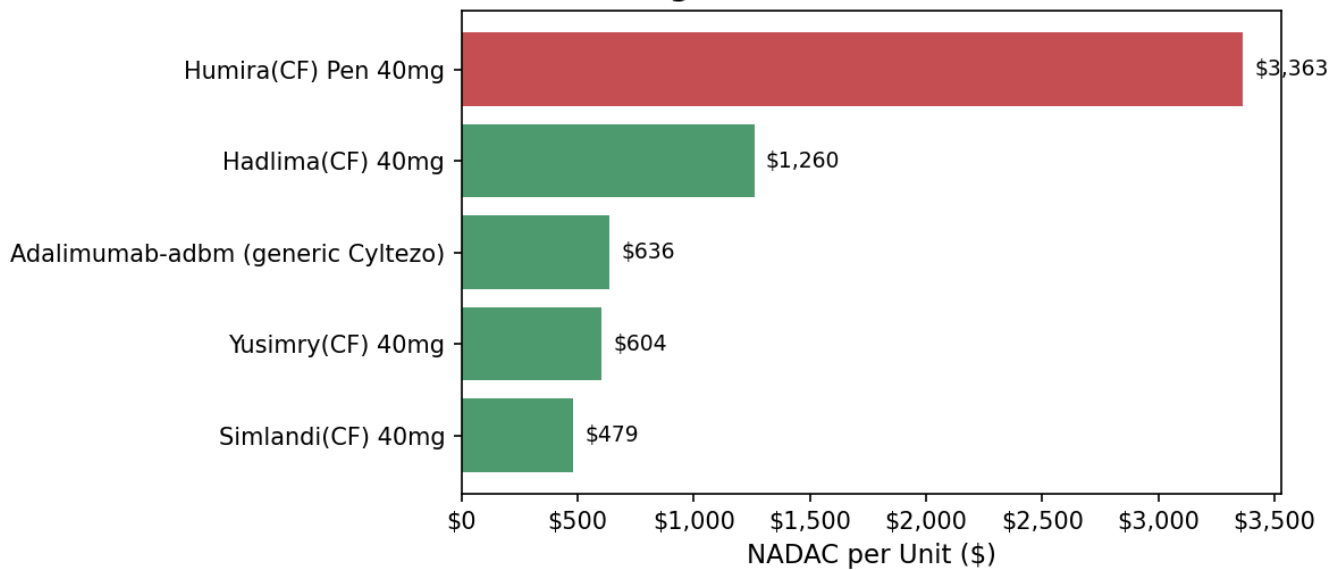
Biosimilars represent <0.4% of NC's total adalimumab utilization in CY2024.

3. NADAC Pricing Comparison

The National Average Drug Acquisition Cost (NADAC) provides a benchmark for pharmacy acquisition costs. The table and chart below compare Humira and its co-preferred biosimilars at the standard 40mg dose. Biosimilar NADAC prices range from \$479 to \$1,260 per unit, representing discounts of 63% to 86% versus Humira.

Product	NADAC/Unit	Discount	Est. Reimb.	Source Date
Humira(CF) Pen 40mg/0.4mL	\$3,362.70	---	\$3,868	2025-06
Simlandi(CF) 40mg	\$479.27	86%	\$551	2025
Yusimry(CF) 40mg	\$604.15	82%	\$695	2025
Adalimumab-adbm (generic Cyltezo)	\$636.04	81%	\$732	2025
Hadlima(CF) 40mg	\$1,260.33	63%	\$1,450	2025
Blended Biosimilar Average	\$744.95	78%	\$857	---

NADAC Pricing: Humira vs. Co-Preferred Biosimilars

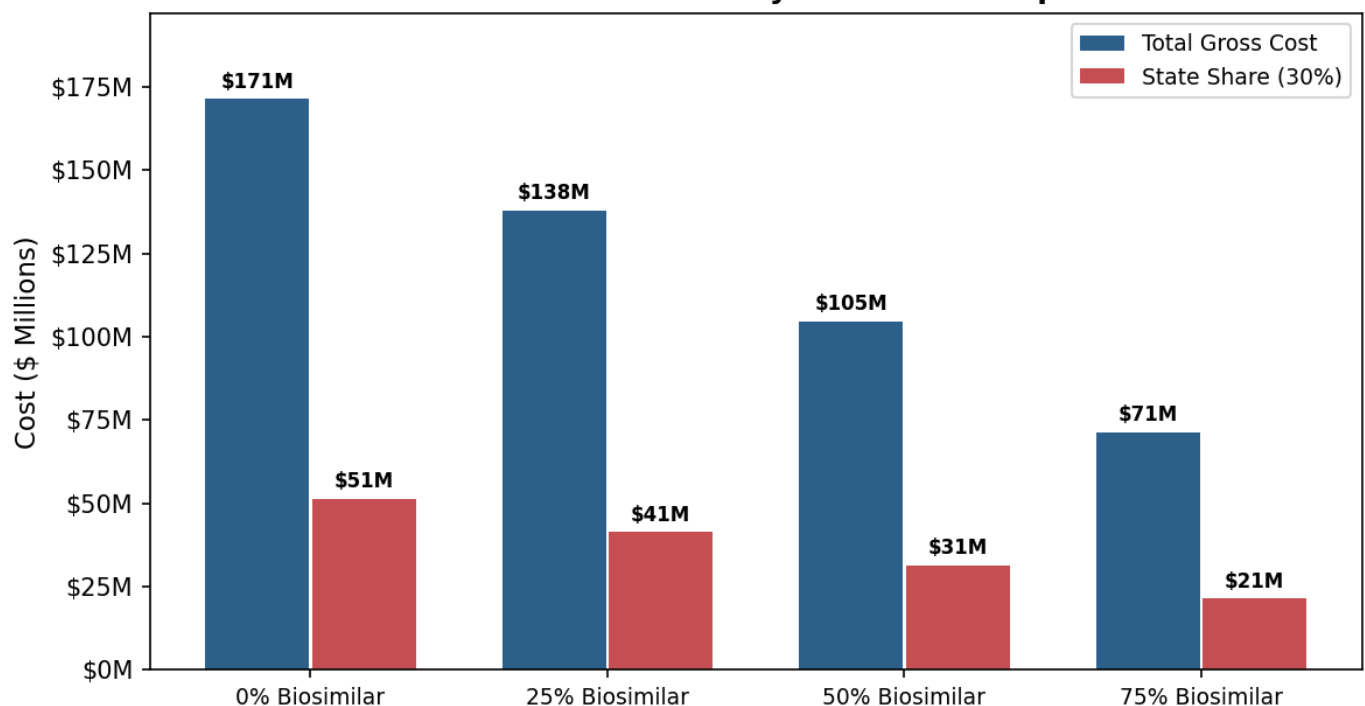


4. Scenario Analysis: Gross Reimbursement Impact

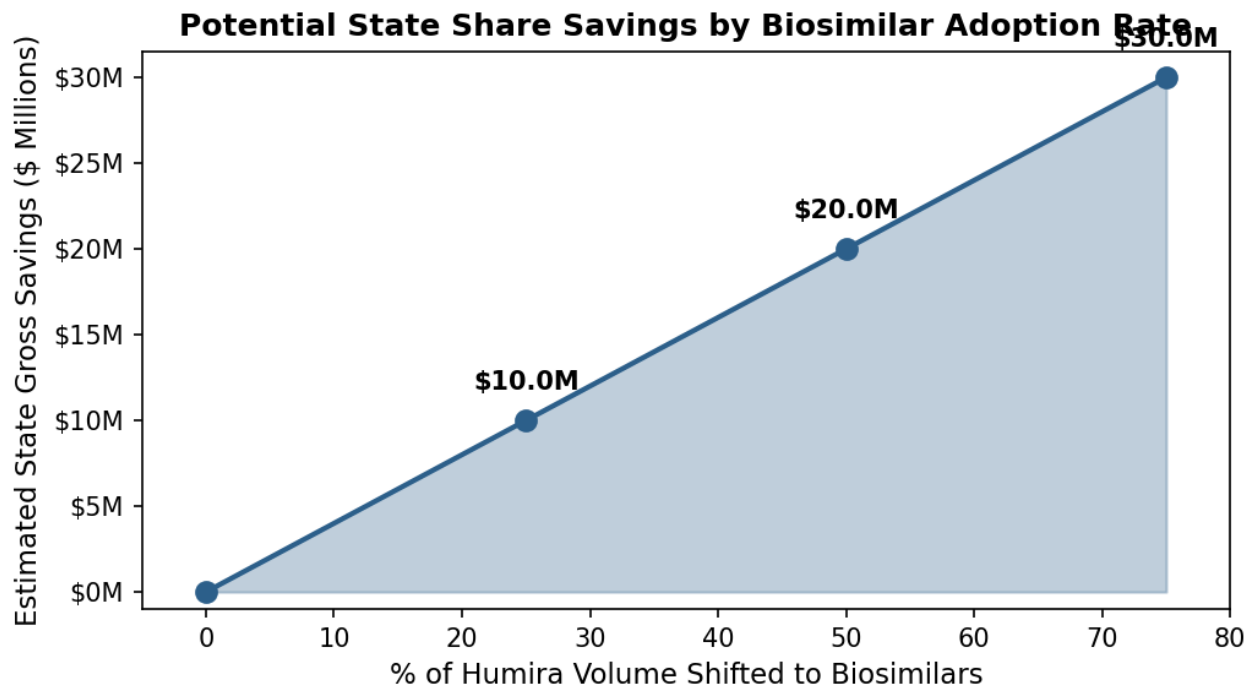
The following scenarios model the gross reimbursement impact of shifting Humira utilization to biosimilars at 0%, 25%, 50%, and 75% adoption rates. Total utilization is held constant at the CY2024 level (44,310 units). The biosimilar reimbursed cost per unit is estimated by applying the NADAC discount ratio to the actual NC Humira reimbursement rate. State share is calculated at 30% (blended FMAP).

Scenario	Bio %	Gross Total	State (30%)	Gross Savings	State Savings	% Sav
0% Biosimilar	0%	\$171.4M	\$51.4M	---	---	---
25% Biosimilar Shift	25%	\$138.0M	\$41.4M	\$33.4M	\$10.0M	19%
50% Biosimilar Shift	50%	\$104.7M	\$31.4M	\$66.7M	\$20.0M	39%
75% Biosimilar Shift	75%	\$71.3M	\$21.4M	\$100.1M	\$30.0M	58%

NC Medicaid Adalimumab Cost by Biosimilar Adoption Scenario



State Share Savings by Adoption Rate



Detailed Scenario Breakdown

0% Biosimilar (Baseline):

Humira: 44,310 units x \$3,868.05 = \$171,393,345

Biosimilar: 0 units x \$856.90 = \$0

Total: \$171,393,345 | State: \$51,418,004 | Savings: \$0

25% Biosimilar Shift:

Humira: 33,232 units x \$3,868.05 = \$128,545,009

Biosimilar: 11,078 units x \$856.90 = \$9,492,301

Total: \$138,037,310 | State: \$41,411,193 | Savings: \$33,356,035

50% Biosimilar Shift:

Humira: 22,155 units x \$3,868.05 = \$85,696,673

Biosimilar: 22,155 units x \$856.90 = \$18,984,602

Total: \$104,681,275 | State: \$31,404,382 | Savings: \$66,712,070

75% Biosimilar Shift:

Humira: 11,078 units x \$3,868.05 = \$42,848,336

Biosimilar: 33,232 units x \$856.90 = \$28,476,903

Total: \$71,325,240 | State: \$21,397,572 | Savings: \$100,068,105

5. Important Caveats: Medicaid Rebate Dynamics

The scenario analysis above reflects gross reimbursement costs. Actual Medicaid net costs are substantially affected by manufacturer rebates under the Medicaid Drug Rebate Program (MDRP). The rebate dynamics between brand and biosimilar products create a well-documented 'rebate trap' that can invert the expected cost relationship.

Humira (Brand, on market since 2002)

- Basic rebate: 23.1% of Average Manufacturer Price (AMP)
- CPI penalty: Likely very large. Humira has been on the market for 20+ years with significant price increases above the Consumer Price Index. The CPI-based additional rebate accumulates over time, potentially pushing the total unit rebate amount to 80-100% of AMP.
- Net cost to Medicaid may approach near-zero after rebates.

Biosimilars (Recently Launched)

- Basic rebate: 13% of AMP (lower statutory rate for biosimilars)
- CPI penalty: Minimal, as these products have been on the market only 1-3 years with limited price inflation.
- Net cost remains close to the gross reimbursement amount.

Implications for NC

Shifting from Humira to biosimilars may increase net Medicaid costs in the short term due to loss of Humira's accumulated CPI rebates. However, NC's PDL decision may reflect: (1) negotiated supplemental rebates with biosimilar manufacturers that offset this gap; (2) a long-term market strategy to increase biosimilar competition and lower overall prices; (3) alignment with federal policy direction supporting biosimilar adoption; or (4) clinical policy goals around access and formulary modernization.

A complete fiscal analysis requires access to North Carolina's confidential supplemental rebate data, which is not available in public CMS datasets.

6. Methodology

Data Sources

1. CMS State Drug Utilization Data (CY2024): Quarterly utilization data reported by state Medicaid agencies to CMS, including units reimbursed, number of prescriptions, and total/Medicaid amounts reimbursed. Dataset ID: 61729e5a-7aa8-448c-8903-ba3e0cd0ea3c. Queried via the data.medicaid.gov DKAN datastore API.
2. NADAC (National Average Drug Acquisition Cost) CY2025: CMS survey-based pricing reflecting the average acquisition cost paid by retail pharmacies. Dataset ID: f38d0706-1239-442c-a3cc-40ef1b686ac0. Used for biosimilar-to-brand price ratio.
3. NC Preferred Drug List: 'NC PDL Effective January 1, 2026 revised 03.18.2026.pdf'. Page 23, Cytokine and CAM Antagonists class, identifies co-preferred biosimilars.

Humira Baseline Calculation

All CMS State Drug Utilization Data records for North Carolina with product_name containing 'HUMIRA' were aggregated across all four quarters of CY2024 and all utilization types (FFSU and MCOU). This includes Humira, Humira(CF), and Humira Pen across all NDCs and strengths. The average reimbursed cost per unit (\$3,868.05) represents the actual blended rate paid by NC Medicaid across all Humira formulations.

Biosimilar Cost Estimation

Because NC had negligible biosimilar utilization in CY2024, the estimated biosimilar cost per unit was derived using NADAC pricing ratios rather than actual reimbursement data. The methodology:

1. Identified the co-preferred biosimilars on the NC PDL with available NADAC data: Simlandi (\$479/unit), Yusimry (\$604/unit), adalimumab-adbm/generic Cyltezo (\$636/unit), and Hadlima (\$1,260/unit).
2. Calculated a simple average (equal-weight) NADAC across these four products: \$744.95/unit.
3. Computed the NADAC ratio: $\$744.95 / \$3,362.70 = 0.2215$.
4. Applied this ratio to the actual NC Humira reimbursed cost per unit: $\$3,868.05 \times 0.2215 = \$856.90/\text{unit}$.

Scenario Modeling

Each scenario holds total adalimumab utilization constant at 44,310 units (CY2024 Humira volume) and varies the proportion filled by biosimilars (0%, 25%, 50%, 75%). Gross reimbursement is calculated as:

$$\text{Total Cost} = (\text{Humira units} \times \$3,868.05) + (\text{Biosimilar units} \times \$856.90)$$

State share is applied at 30% (user-specified blended FMAP for NC).

Limitations

1. Gross costs only: Does not account for MDRP basic, CPI, or supplemental rebates. Net cost impact may differ substantially and could potentially be inverse.
2. Static utilization: Assumes total adalimumab volume remains constant. Policy changes could increase or decrease total utilization.
3. Equal-weight biosimilar pricing: Uses a simple average across four biosimilars. Actual market share distribution

among biosimilars will affect the realized discount.

4. NADAC-based estimation: Applies NADAC price ratios to estimate biosimilar reimbursement. Actual reimbursement rates will vary based on state-specific methodologies, dispensing fees, and contract terms.

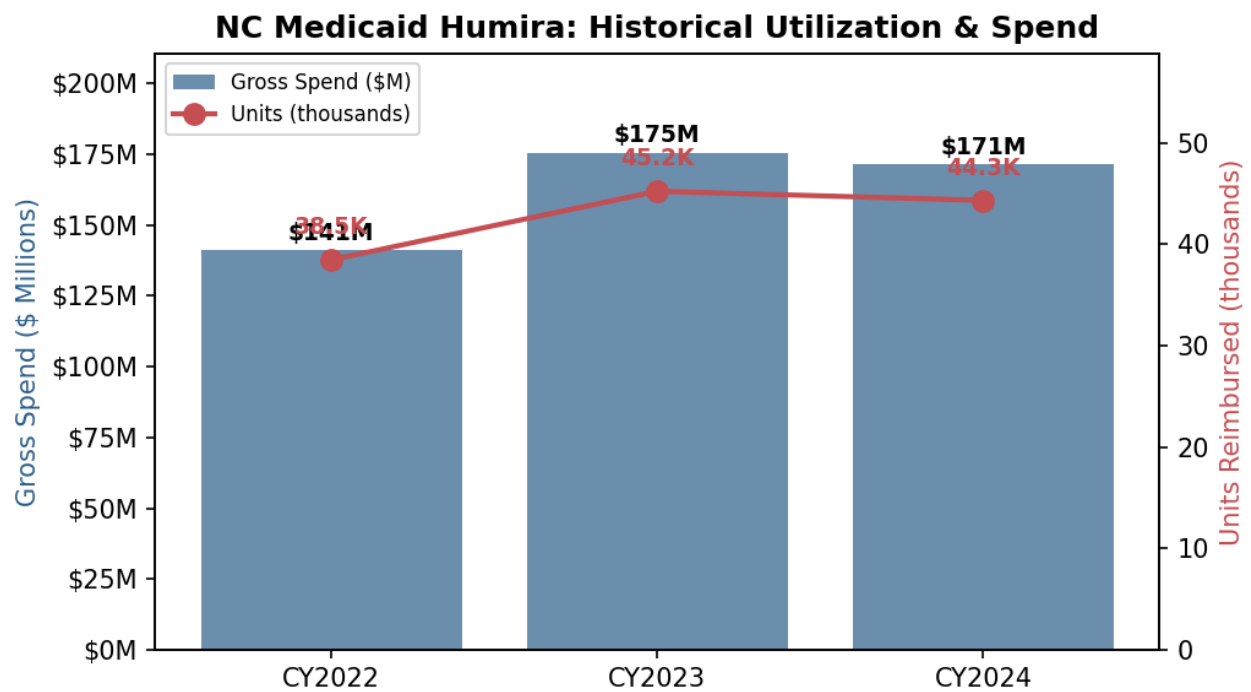
5. Data lag: CMS SDU data for CY2024 may not reflect the most recent quarter's revisions. NADAC rates are point-in-time and subject to weekly updates.

6. Excludes managed care: MCOU records reflect managed care utilization but the rebate and cost-sharing structures within MCOs may differ from FFS.

7. Historical Utilization Trend (CY2022-CY2024)

NC Medicaid Humira utilization grew significantly from CY2022 to CY2023 (+17.6% in units, +24.3% in gross spend), then stabilized in CY2024 (-2.0% in units, -2.3% in spend). Cost per unit has been relatively flat since CY2023 (~\$3,870-\$3,880). This trend informs the projection assumption of flat utilization and modest (~2%) annual price inflation.

Year	Units	Rx	Gross Total	Medicaid	Avg/Unit	YoY Chg
CY2022	38,450	14,665	\$141.1M	\$139.9M	\$3,671	---
CY2023	45,213	16,482	\$175.4M	\$173.0M	\$3,880	+24.3%
CY2024	44,310	16,238	\$171.4M	\$167.1M	\$3,868	-2.3%



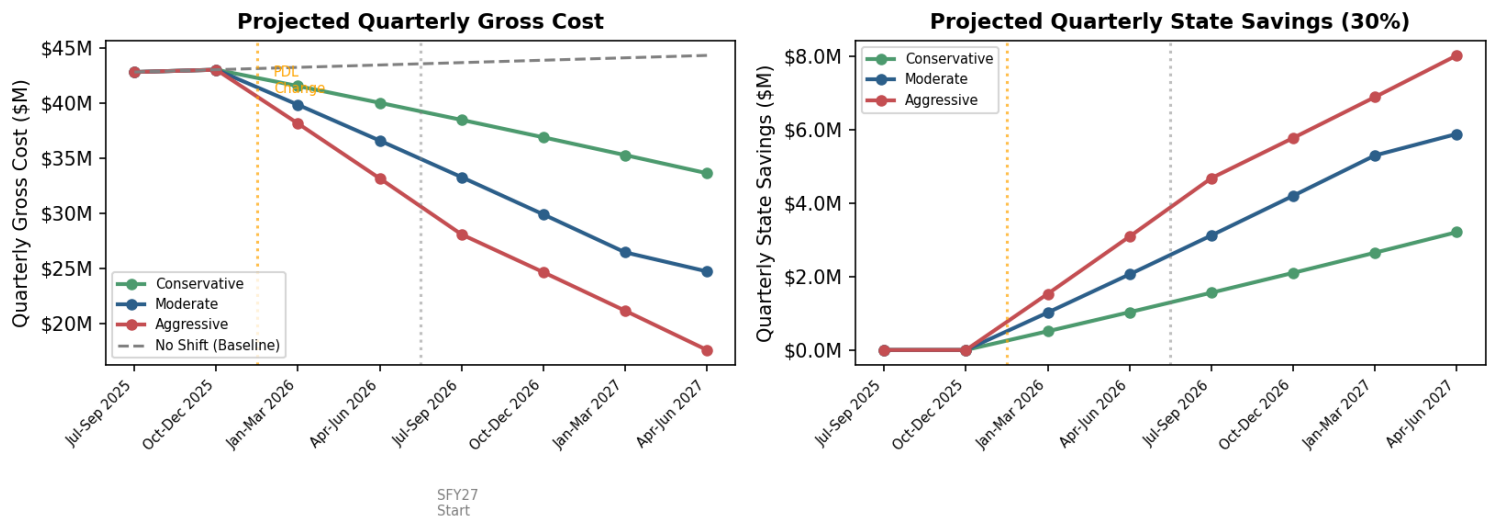
8. SFY26-SFY27 Cost Savings Projection

The following projections model gross cost savings through SFY27 (ending June 30, 2027) under three biosimilar adoption scenarios. The NC PDL change takes effect January 1, 2026, which falls in the second half of SFY26. Three adoption ramp scenarios reflect varying speed of biosimilar uptake after the PDL change:

- Conservative: Slow adoption, reaching 30% biosimilar share by end of SFY27
- Moderate: Steady adoption, reaching 55% by end of SFY27
- Aggressive: Rapid adoption, reaching 75% by end of SFY27

Projection Assumptions

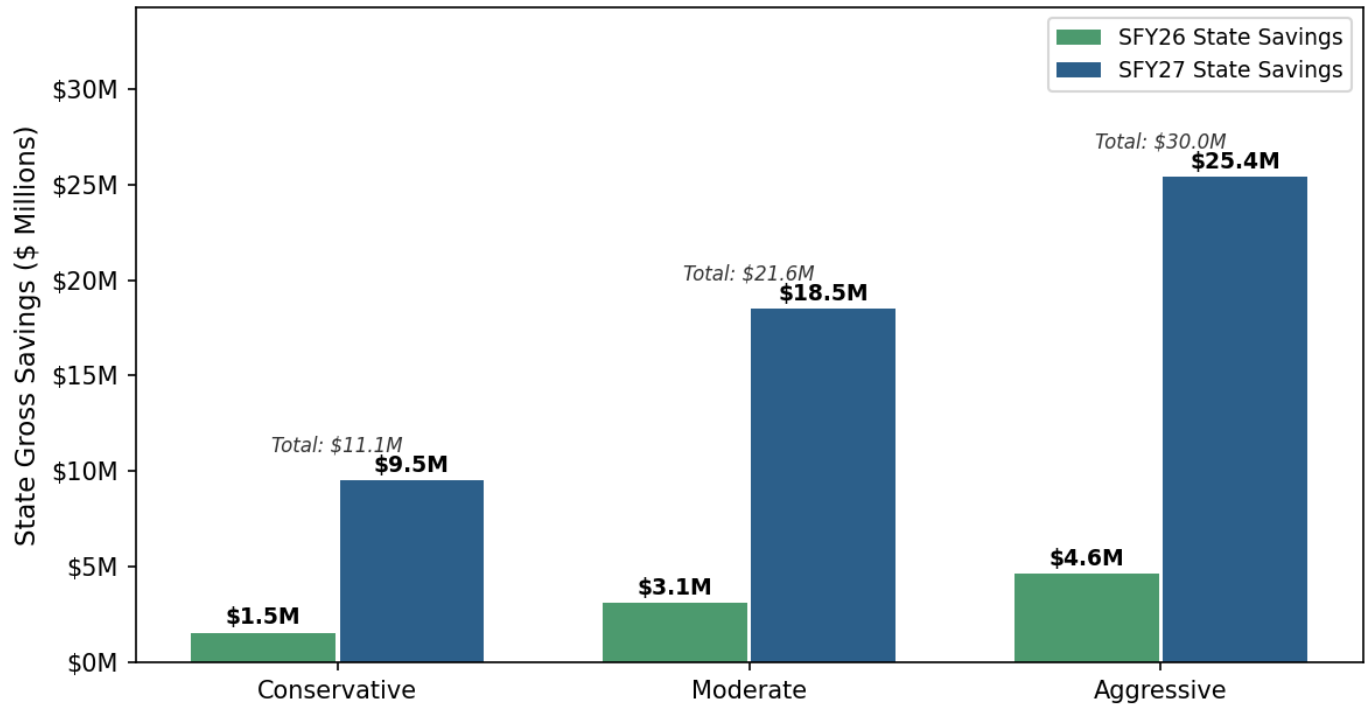
Base quarterly utilization	11,078 units (CY2024 annualized)
Humira annual price inflation	2%
Biosimilar annual price deflation	5% (competitive erosion)
PDL change effective	January 1, 2026 (SFY26 Q3)
NC State Fiscal Year	July 1 - June 30
State share	30% (blended FMAP)



SFY Annual Summary by Scenario

Scenario	SFY26 Gross	SFY26 State	SFY27 Gross	SFY27 State	Cumul. State	
Conservative	\$1.5M	\$1.5M	\$9.5M	\$9.5M	\$11.1M	
Moderate	\$3.1M	\$3.1M	\$18.5M	\$18.5M	\$21.6M	
Aggressive	\$4.6M	\$4.6M	\$25.4M	\$25.4M	\$30.0M	
No Shift (Base)	\$172.7M	---	\$176.1M	---	---	

Cumulative State Savings by Adoption Scenario (SFY26 + SFY27)



Quarterly Detail: Conservative Adoption Scenario

Quarterly biosimilar share ramp: 0%, 0%, 5%, 10%, 15%, 20%, 25%, 30%

Quarter	Bio%	Humira \$	Biosim \$	Gross Total	Baseline	Savings	State Sav
Jul-Sep 2025	0%	\$42.8M	\$0.0M	\$42.8M	\$42.8M	\$0.0M	\$0.0M
Oct-Dec 2025	0%	\$43.1M	\$0.0M	\$43.1M	\$43.1M	\$0.0M	\$0.0M
Jan-Mar 2026	5%	\$41.1M	\$0.5M	\$41.6M	\$43.3M	\$1.7M	\$0.5M
Apr-Jun 2026	10%	\$39.1M	\$0.9M	\$40.1M	\$43.5M	\$3.4M	\$1.0M
SFY26 Total				\$167.5M	\$172.7M	\$5.1M	\$1.5M
Jul-Sep 2026	15%	\$37.1M	\$1.4M	\$38.5M	\$43.7M	\$5.2M	\$1.6M
Oct-Dec 2026	20%	\$35.1M	\$1.8M	\$36.9M	\$43.9M	\$7.0M	\$2.1M
Jan-Mar 2027	25%	\$33.1M	\$2.2M	\$35.3M	\$44.1M	\$8.8M	\$2.7M
Apr-Jun 2027	30%	\$31.1M	\$2.6M	\$33.7M	\$44.4M	\$10.7M	\$3.2M
SFY27 Total				\$144.4M	\$176.1M	\$31.7M	\$9.5M

Cumulative State Savings (SFY26+SFY27): \$11.1M (10.6% of baseline)

Quarterly Detail: Moderate Adoption Scenario

Quarterly biosimilar share ramp: 0%, 0%, 10%, 20%, 30%, 40%, 50%, 55%

Quarter	Bio%	Humira \$	Biosim \$	Gross Total	Baseline	Savings	State Sav
Jul-Sep 2025	0%	\$42.8M	\$0.0M	\$42.8M	\$42.8M	\$0.0M	\$0.0M
Oct-Dec 2025	0%	\$43.1M	\$0.0M	\$43.1M	\$43.1M	\$0.0M	\$0.0M
Jan-Mar 2026	10%	\$38.9M	\$0.9M	\$39.9M	\$43.3M	\$3.4M	\$1.0M
Apr-Jun 2026	20%	\$34.8M	\$1.8M	\$36.6M	\$43.5M	\$6.9M	\$2.1M
SFY26 Total				\$162.4M	\$172.7M	\$10.3M	\$3.1M
Jul-Sep 2026	30%	\$30.6M	\$2.7M	\$33.3M	\$43.7M	\$10.4M	\$3.1M
Oct-Dec 2026	40%	\$26.4M	\$3.6M	\$29.9M	\$43.9M	\$14.0M	\$4.2M
Jan-Mar 2027	50%	\$22.1M	\$4.4M	\$26.5M	\$44.1M	\$17.7M	\$5.3M
Apr-Jun 2027	55%	\$20.0M	\$4.8M	\$24.7M	\$44.4M	\$19.6M	\$5.9M
SFY27 Total				\$114.4M	\$176.1M	\$61.7M	\$18.5M

Cumulative State Savings (SFY26+SFY27): \$21.6M (20.6% of baseline)

Quarterly Detail: Aggressive Adoption Scenario

Quarterly biosimilar share ramp: 0%, 0%, 15%, 30%, 45%, 55%, 65%, 75%

Quarter	Bio%	Humira \$	Biosim \$	Gross Total	Baseline	Savings	State Sav
Jul-Sep 2025	0%	\$42.8M	\$0.0M	\$42.8M	\$42.8M	\$0.0M	\$0.0M
Oct-Dec 2025	0%	\$43.1M	\$0.0M	\$43.1M	\$43.1M	\$0.0M	\$0.0M
Jan-Mar 2026	15%	\$36.8M	\$1.4M	\$38.2M	\$43.3M	\$5.1M	\$1.5M
Apr-Jun 2026	30%	\$30.4M	\$2.7M	\$33.2M	\$43.5M	\$10.3M	\$3.1M
SFY26 Total				\$157.3M	\$172.7M	\$15.4M	\$4.6M
Jul-Sep 2026	45%	\$24.0M	\$4.1M	\$28.1M	\$43.7M	\$15.6M	\$4.7M
Oct-Dec 2026	55%	\$19.8M	\$4.9M	\$24.7M	\$43.9M	\$19.3M	\$5.8M
Jan-Mar 2027	65%	\$15.4M	\$5.7M	\$21.2M	\$44.1M	\$23.0M	\$6.9M
Apr-Jun 2027	75%	\$11.1M	\$6.5M	\$17.6M	\$44.4M	\$26.8M	\$8.0M
SFY27 Total				\$91.5M	\$176.1M	\$84.6M	\$25.4M

Cumulative State Savings (SFY26+SFY27): \$30.0M (28.7% of baseline)

9. SFY Projection Methodology

The SFY26-SFY27 projections extend the point-in-time scenario analysis into a time-series model that accounts for the PDL change timeline, gradual biosimilar adoption, and price dynamics.

Adoption Curve Rationale

Biosimilar adoption following a PDL co-preferred designation typically follows an S-curve pattern. New starts shift first, while existing patients on Humira convert more gradually. The three scenarios bracket the range of plausible outcomes:

- Conservative (30% by SFY27 end): Reflects provider inertia, patient reluctance to switch, and limited plan-level enforcement of biosimilar preference.
- Moderate (55% by SFY27 end): Consistent with states that have implemented co-preferred status with active utilization management (step therapy, prior authorization encouraging biosimilar starts).
- Aggressive (75% by SFY27 end): Assumes strong plan-level enforcement, mandatory biosimilar-first policies for new starts, and active switching programs for stable patients.

Price Dynamics

Humira's cost per unit is inflated at 2% annually, consistent with the modest year-over-year increases observed in CY2022-CY2024 NC data (CY2022: \$3,671 -> CY2024: \$3,868, ~2.7% CAGR).

Biosimilar cost per unit is deflated at 5% annually, reflecting expected competitive pricing pressure as multiple biosimilars compete for market share following the PDL change. This is conservative relative to the 10-30% annual price erosion observed in mature biosimilar markets (e.g., infliximab biosimilars).

Key Projection Limitations

1. Adoption ramps are illustrative, not predictive. Actual adoption depends on plan-level policies, provider behavior, patient preferences, and manufacturer contracting.
2. Utilization volume is held flat. In practice, lower costs may increase utilization (demand elasticity), and the expanded preferred list may attract patients from competing biologics (e.g., etanercept).
3. Rebate dynamics (Section 5) apply to projections as well. Projected gross savings may not translate to net savings without supplemental rebate offsets.
4. MCO contracting: NC Medicaid managed care organizations may negotiate separate rebate and formulary arrangements that diverge from the state PDL.